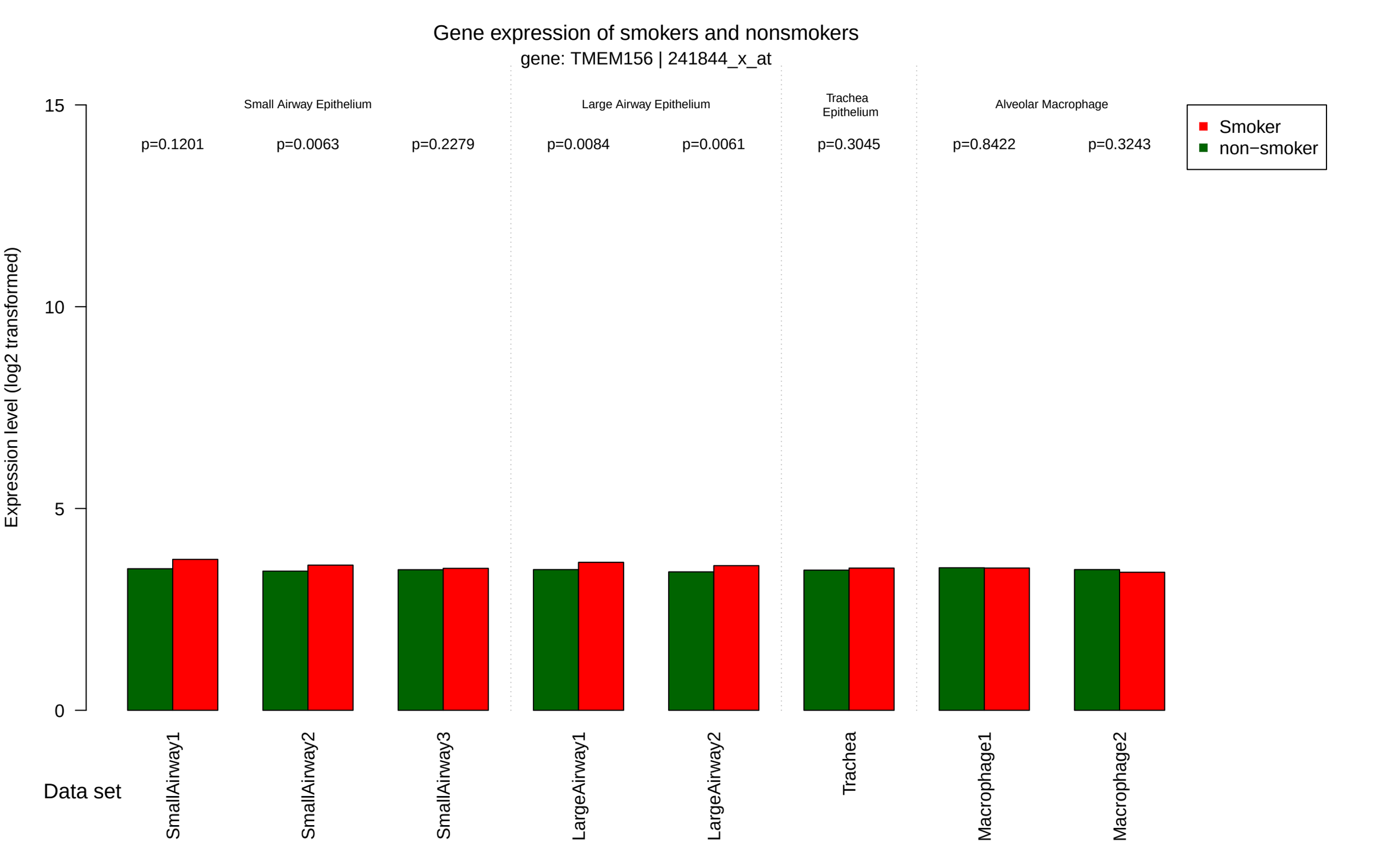
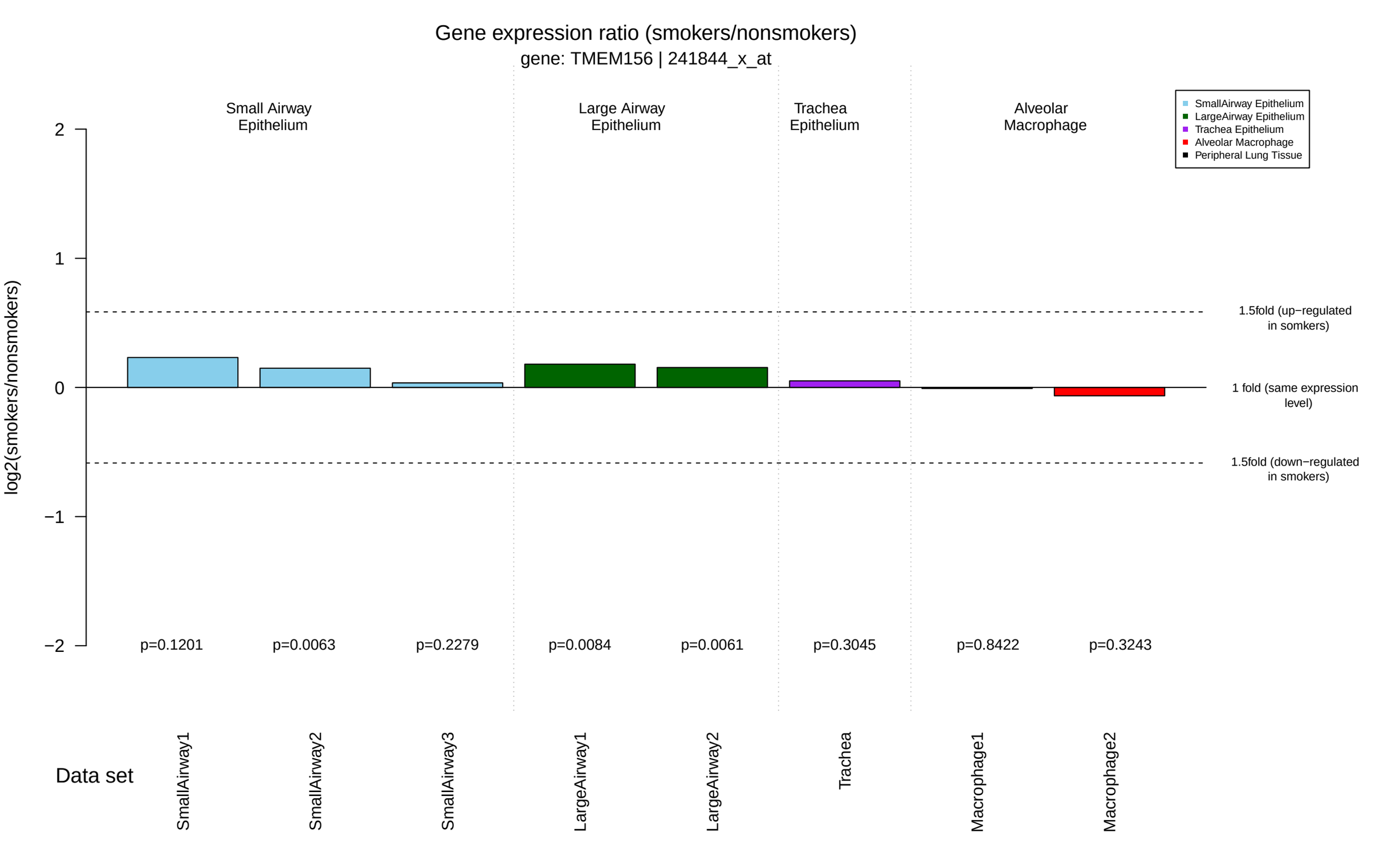
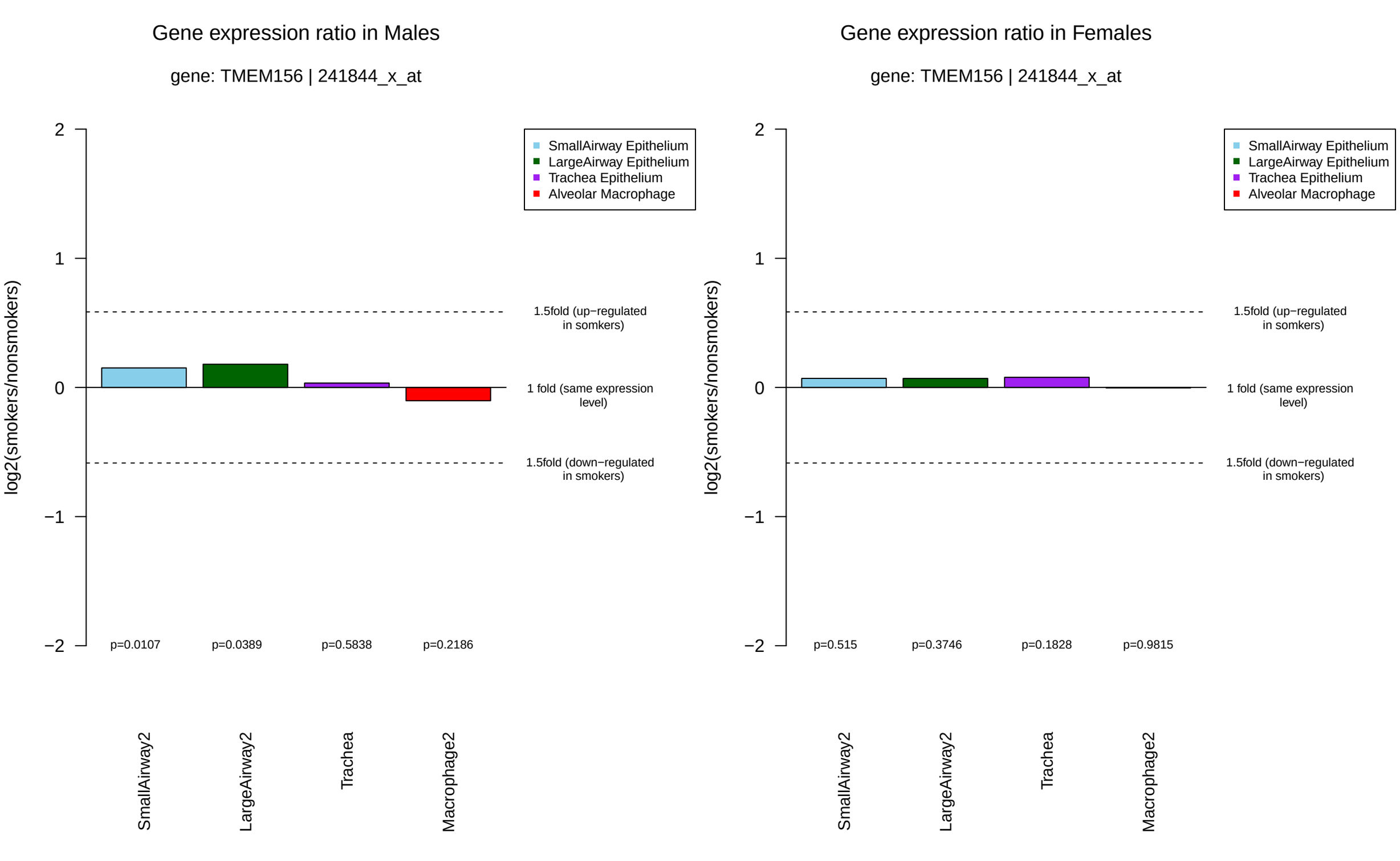
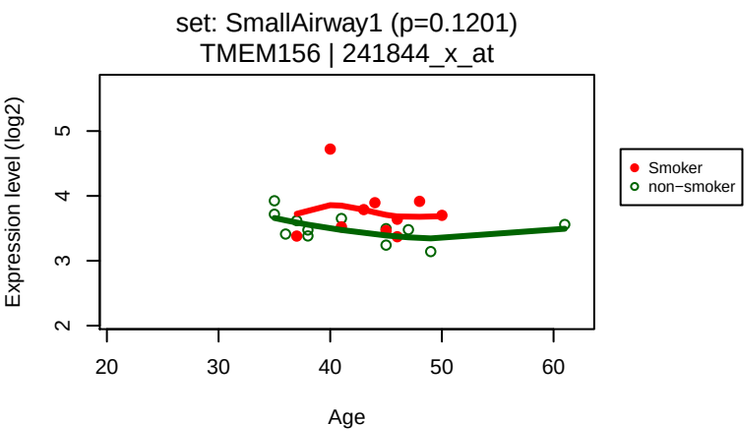
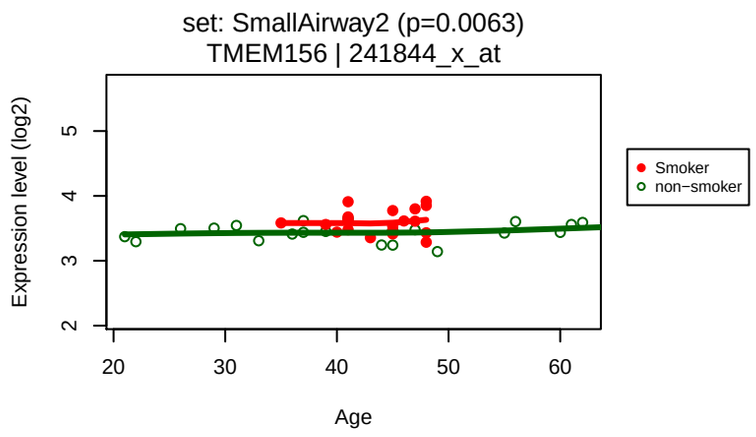
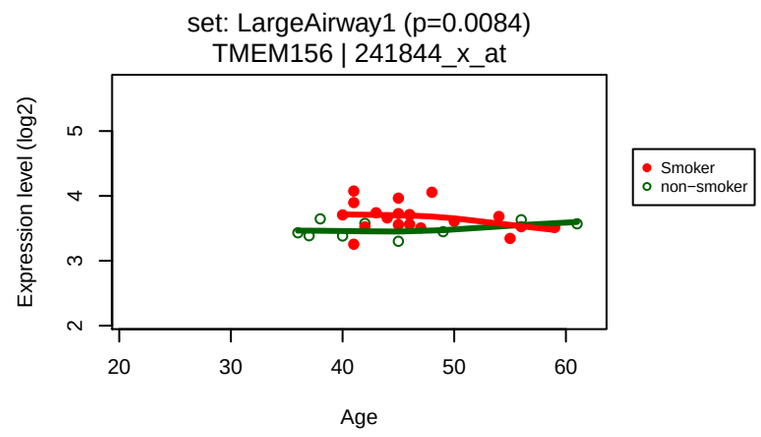

Gene expression of smokers and nonsmokers
gene: TMEM156 | 241844_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1201
p=0.0063
p=0.2279
p=0.0084
p=0.0061
p=0.3045
p=0.8422
p=0.3243
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

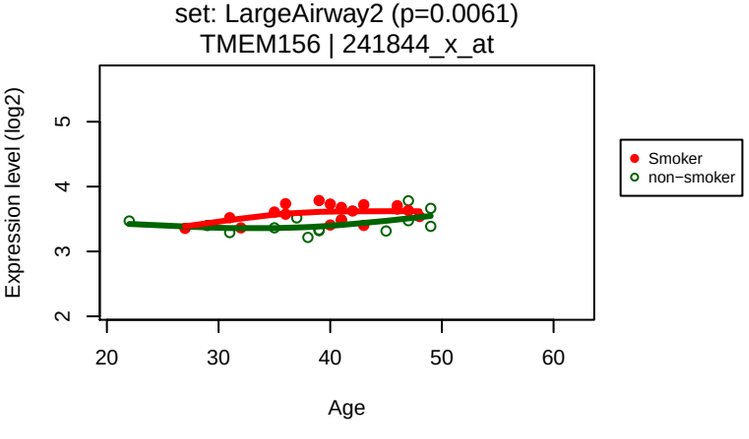

Gene expression ratio (smokers/nonsmokers)
gene: TMEM156 | 241844_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1201
p=0.0063
p=0.2279
p=0.0084
p=0.0061
p=0.3045
p=0.8422
p=0.3243
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

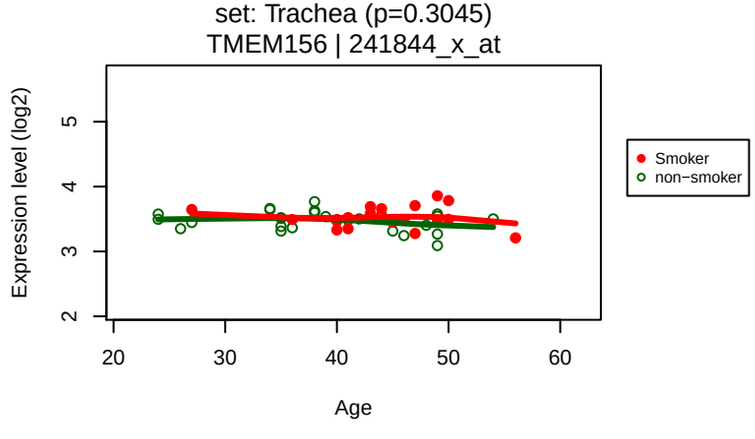

Gene expression ratio in Males
gene: TMEM156 | 241844_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0107
p=0.0389
p=0.5838
p=0.2186
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM156 | 241844_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.515
p=0.3746
p=0.1828
p=0.9815
SmallAirway2
LargeAirway2
Trachea
Macrophage2

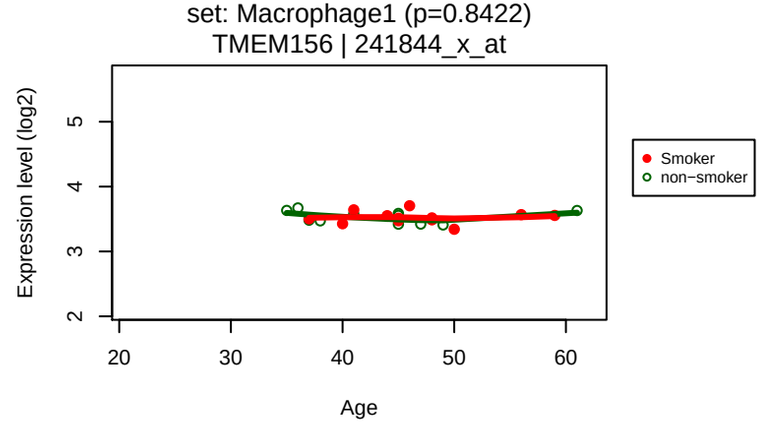

set: SmallAirway1 (p=0.1201)
TMEM156 | 241844_x_at
Expression level (log2)
Age
Smoker
non−smoker

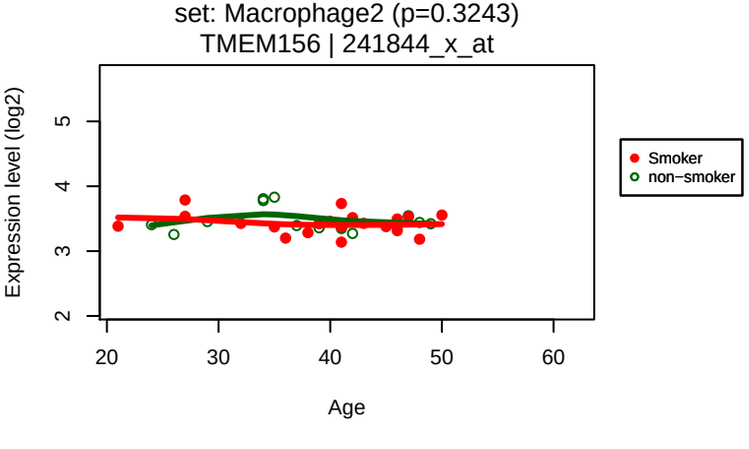

set: SmallAirway2 (p=0.0063)
TMEM156 | 241844_x_at
Expression level (log2)
Age
Smoker
non−smoker

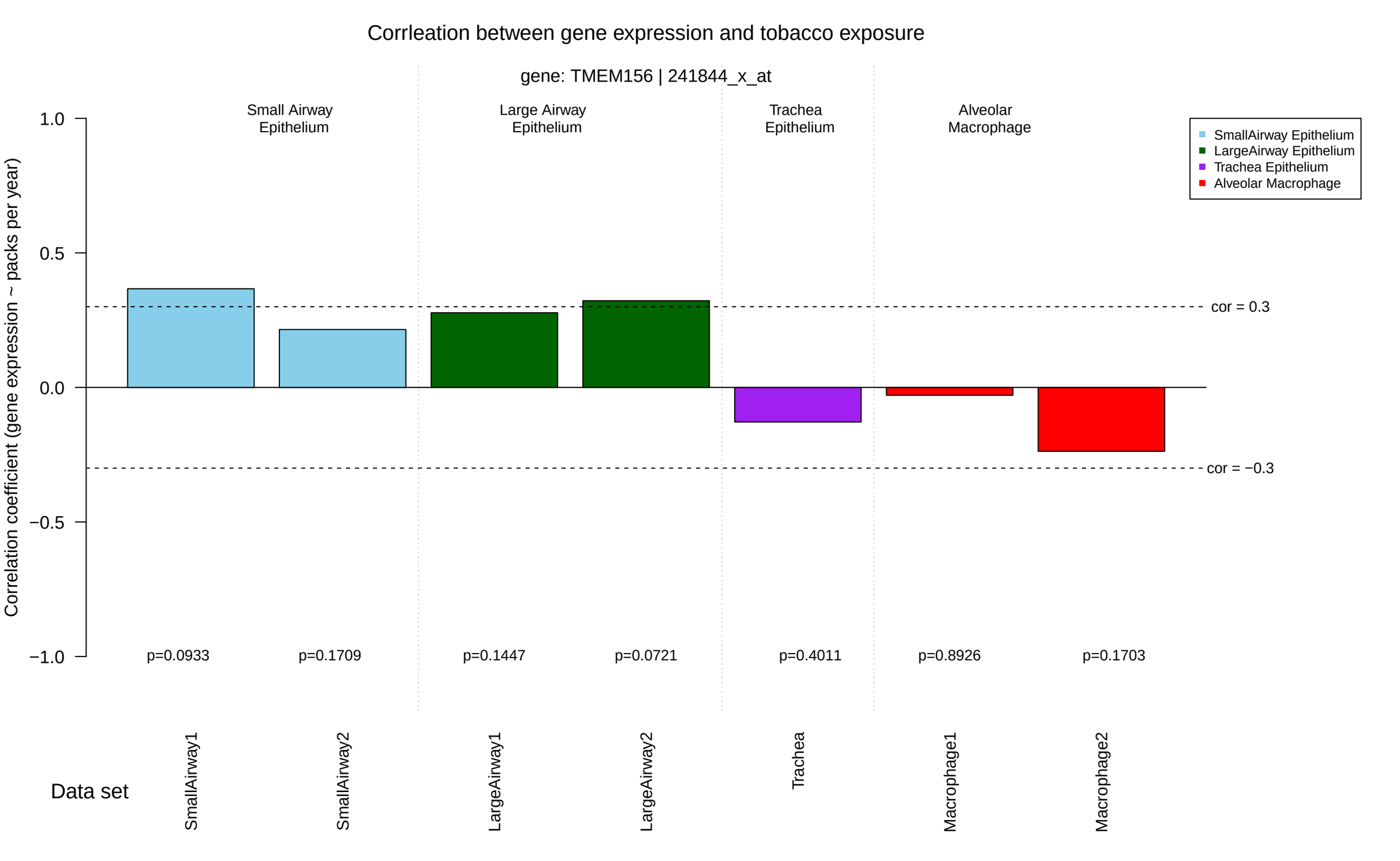

set: LargeAirway1 (p=0.0084)
TMEM156 | 241844_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0061)
TMEM156 | 241844_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3045)
TMEM156 | 241844_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8422)
TMEM156 | 241844_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3243)
TMEM156 | 241844_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM156 | 241844_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0933
p=0.1709
p=0.1447
p=0.0721
p=0.4011
p=0.8926
p=0.1703
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2